



PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended pelvic lymph node dissection by preoperative nomograms

Qiaoke Ma¹ · Bei Chen¹ · Robert Seifert² · Rui Zhou³ · Ling Xiao¹ · Jinhui Yang¹ · Axel Rominger² · Kuangyu Shi^{2,4} · Weikai Li⁵ · Yongxiang Tang^{1,2,6} · Shuo Hu^{1,6,7}

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Abstract

Purpose To develop and validate a prostate-specific membrane antigen (PSMA) PET/CT based multimodal deep learning model for predicting pathological lymph node invasion (LNI) in prostate cancer (PCa) patients identified as candidates for extended pelvic lymph node dissection (ePLND) by preoperative nomograms.

Methods [⁶⁸Ga]Ga-PSMA-617 PET/CT scan of 116 eligible PCa patients (82 in the training cohort and 34 in the test cohort) who underwent radical prostatectomy with ePLND were analyzed in our study. The Med3D deep learning network was utilized to extract discriminative features from the entire prostate volume of interest on the PET/CT images. Subsequently, a multimodal model i.e., Multi kernel Support Vector Machine was constructed to combine the PET/CT deep learning features, quantitative PET and clinical parameters. The performance of the multimodal models was assessed using final histopathology as the reference standard, with evaluation metrics including area under the receiver operating characteristic curve (AUC), calibration curve, decision curve analysis, and compared with available nomograms and PET/CT visual evaluation result.

Results Our multimodal model incorporated clinical information, maximum standardized uptake value (SUV_{max}), and PET/CT deep learning features. The AUC for predicting LNI was 0.89 (95% confidence interval [CI] 0.81–0.97) for the final model. The proposed model demonstrated superior predictive accuracy in the test cohort compared to PET/CT visual evaluation result, the Memorial Sloan Kettering Cancer Center (MSKCC) and the Briganti-2017 nomograms (AUC 0.85 [95% CI 0.69–1.00] vs. 0.80 [95% CI 0.64–0.95] vs. 0.79 [95% CI 0.61–0.97] and 0.69 [95% CI 0.50–0.88], respectively). The proposed model showed similar calibration and higher net benefit as compared to the traditional nomograms.

Conclusion Our multimodal deep learning model, which incorporates preoperative PSMA PET/CT imaging, shows enhanced predictive capabilities for LNI in clinically localized PCa compared to PSMA PET/CT visual evaluation result and existing nomograms like the MSKCC and Briganti-2017 nomograms. This model has the potential to reduce unnecessary ePLND procedures while minimizing the risk of missing cases of LNI.

Keywords Prostate cancer · Lymph node invasion · Prostate-specific membrane antigen positron emission tomography · Deep learning

Qiaoke Ma and Bei Chen contributed equally to this work as co-first authors.

Extended author information available on the last page of the article

Abbreviations

ADT	Androgen deprivation therapy
AUC	Area under the receiver operating characteristic curve
CI	Confidence interval
DCA	Decision curve analysis
ePLND	Extended pelvic lymph node dissection
IQR	Interquartile range
ISUP	International Society of Urological Pathology
LNI	Lymph node invasion
mpMRI	Multiparametric magnetic resonance imaging
MSKCC	Memorial Sloan Kettering Cancer Center
NPV	Negative predictive value
PCa	Prostate cancer
PET/CT	Positron emission tomography/computed tomography
PPV	Positive predictive value
PSA	Prostate-specific antigen
PSMA	Prostate-specific membrane antigen
PSMA-RADS	Prostate-specific Membrane Antigen Reporting and Data System
ROC	Receiver operating characteristic curve
RP	Radical prostatectomy
SUV _{max}	Maximum standardized uptake value
SVM	Support vector machine
VOI	Volume of interest

Introduction

For patients with clinically significant and localized prostate cancer (PCa), the extended pelvic lymph node dissection (ePLND) performed during radical prostatectomy (RP) is widely considered the most accurate modality for nodal staging. It plays a vital role in assessing prognosis and identifying patients who may benefit from additional or salvage therapies [1–3]. However, the role of ePLND is still debated. Extended dissecting scope could lead to worse intraoperative and perioperative outcomes, prolonging operation time and increasing costs. Moreover, randomized trials have failed to show a clear benefit in terms of oncologic control, and the procedure is associated with potential complications, such as higher risk of lymphocele, lymph edema and thromboembolic events [4, 5]. Therefore, efforts should be made to optimize this procedure to reduce operative time and the occurrence of surgery-related complications without compromising identification of lymph node invasion (LNI).

According to the guidelines, estimation of an individual's risk of LNI is typically performed using one of the numerous clinical nomograms [6]. With the cutoff percentage

above 2–9%, an ePLND is recommended based on the clinical, biochemical, and pathological preoperative variables. However, commonly used nomograms such as the Memorial Sloan Kettering Cancer Center (MSKCC) nomogram [7] and the Briganti-2017 nomogram [8] have shown to result in unnecessary ePLND in approximately 66–77% of patients above the cutoff according to previous research [8, 9]. Even the Briganti-2019 nomogram, which incorporates imaging findings and targeted biopsy histology following multiparametric magnetic resonance imaging (mpMRI), can lead to a high false positive rate of 31% [9].

Molecular imaging using prostate-specific membrane antigen (PSMA) positron emission tomography (PET) has demonstrated superior accuracy in detecting metastatic lesions in PCa than conventional imaging techniques such as bone scintigraphy, computed tomography (CT), and mpMRI. It is recommended for the primary staging of PCa [6, 10, 11]. Despite its excellent specificity, the spatial resolution of PET limits its detection rate for micro-metastasis, the sensitivity of PSMA PET in identifying LNI disease is suboptimal ranging from 40 to 77% in various studies [10, 12–14], indicating a high negative predictive value for nodal metastases. Currently, PSMA PET is not incorporated into the aforementioned nomograms.

Considering the well-documented correlation between PSMA expression, risk groups and Gleason score [15–17], it is plausible to propose that PSMA uptake in the primary tumor could serve as a predictor for distant disease [18]. Our research team has recently optimized lymph node staging in PCa patients by integrating [⁶⁸Ga]Ga-PSMA PET-derived maximum standardized uptake value (SUV_{max}), neutrophil-lymphocyte ratio, and platelet-lymphocyte ratio [19]. Given that multifocality is present in around 63% of PCa cases [20, 21], it is possible that a small clone within the primary tumor or/and prostate volume with higher aggressiveness may be the origin of metastatic lesions. Therefore, a comprehensive assessment of the entire prostate volume using PSMA PET imaging and artificial intelligence could potentially improve the prediction of LNI in PCa [22–25]. Previous studies have developed and validated an automated convolutional neural network method for detecting LNI using PSMA PET imaging, reporting sensitivity rates in the range of 79–90% [26–28]. However, the patient cohorts employed in these studies lacked histopathological confirmation of lymph node metastases and included a substantial number of patients with bone metastases who had no surgical treatment option. Therefore, the aim of our study was to develop and validate a multimodal deep learning model based on preoperative PSMA PET/CT imaging to accurately predict LNI in a cohort of PCa patients identified as candidates for ePLND by clinical nomograms, and evaluate the added value of deep learning utilizing PSMA PET/CT in

comparison with existing clinical nomograms for assessing the risk of nodal disease.

Materials and methods

Study population

This retrospective study included all PCa patients that underwent [^{68}Ga]Ga-PSMA-617 PET/CT scan for staging prior to RP and ePLND between April 2020 and September 2024 in Xiangya hospital (Changsha, China) and they were divided into the training cohort and the test cohort in a ratio of 7:3. Three available preoperative nomograms were used to determine candidates for ePLND in our hospital. The Briganti-2017 and MSKCC nomograms were used for patients who did not undergo preoperative MRI-targeted biopsies or who had tumor-negative MRI-targeted biopsies [8, 29]. The Briganti-2019 nomogram was applied in patients who had tumor-positive MRI-targeted biopsies of suspicious lesions (Prostate Imaging Reporting and Data System [PI-RADS] 3–5) on mpMRI [9].

The exclusion criteria were defined as follows: (1) any treatment before PET/CT scan including androgen deprivation therapy (ADT), local radiotherapy and any other approach, (2) incomplete clinical or pathological data, (3) interval over a month between PET/CT scan and surgery. This study was approved by the Institutional Review Board of Xiangya Hospital, Central South University and the requirement for written informed consent was waived.

Data collection

From all eligible patients, preoperative clinical, biochemical, radiological, and histopathological data were obtained. Clinical parameters: age, initial prostate-specific antigen (PSA) at the time of PSMA PET, the highest International Society of Urological Pathology (ISUP) grade group (grade groups 1–5) [30] at systematic biopsy and clinical tumor stage (cT stage) based on mpMRI and PSMA PET/CT imaging.

All PSMA PET/CT scans were reviewed by two nuclear medicine physicians (QK Ma and B Chen) following the Prostate-specific Membrane Antigen Reporting and Data System (PSMA-RADS) V2.0 guideline [31] and qualitative PET/CT assessment for lymph node metastasis was done based on visual evaluation (miN0 representing LNI negative; miN1 representing LNI positive). A PSMA-RADS score ranging from 1 to 5 was given to every visible pelvic lymph node on PSMA PET/CT images by each reader respectively. PSMA-RADS 1–3 lesion was considered benign, while PSMA-RADS 4–5 was considered malignant

in our study. Care was taken to avoid LNI false-positive findings by considering physiological biodistribution and the known PSMA positive pitfalls such as neural ganglia, Paget's disease, sarcoidosis and others. Besides, SUV_{max} as a semi-quantitative [^{68}Ga]Ga-PSMA-617 PET parameter of the primary tumor was obtained. Any focal tracer uptake in the prostate visibly higher than the surrounding normal prostate tissue or with a 2–3 PSMA expression score as recommended by PROMISE V2 framework [32], and not attributed to physiological accumulation or benign entity was considered possible malignancy and documented. In cases where the two nuclear medicine physicians had disagreements regarding the final determination of lymph node metastasis as well as localization of the primary tumor and SUV_{max} measurement, a consensus was reached through discussion. When a consensus could not be reached, a third experienced senior nuclear medicine specialist (S Hu) reviewed the case and make the final decision.

Other histopathological information required in common nomograms including percentage of positive cores in systematic biopsy and percentage of positive cores with the highest-grade/lower-grade disease was also obtained. Moreover, pathological lymph node status diagnosed by experienced pathologists was collected from the postoperative pathological report of every patient and was considered as the reference standard of our study.

[^{68}Ga]Ga-PSMA-617 PET/CT imaging procedure

All patients received an intravenous injection of [^{68}Ga]Ga-PSMA-617 (median dose, 160.6 MBq; range, 146.5–177.9 MBq) and PET/CT images were acquired 45–60 min after using a General Electric Discovery PET/CT 690 Elite scanner (GE, Healthcare). After low-dose non-contrast-enhanced CT scan (parameters: 120 kV; automatic mAs; spacing: 1:1; layer thickness: 3.75 mm; matrix: 512×512), 3D emission-mode PET acquisitions were then obtained from the top of the skull to the middle of the thighs for 5–7 bed positions at 2 min per position with a layer thickness of 3.27 mm. Reconstruction of PET data by an ordered-subsets expectation maximization algorithm using 2 iterations and 23 subsets.

Extended pelvic lymph-node dissection and histopathological examination

All curative surgery procedures were performed by experienced urologists in our institution in form of either a laparoscopic RP or a robot-assisted laparoscopic RP with bilateral ePLND. Bilateral ePLND involved removing the lymph nodes located lateral to the internal iliac vessels and within the obturator fossa, extending up to and including

the crossing between the ureter and common iliac vessels, along the external iliac vessels [33]. Lymph-node specimens were examined by high-volume, committed uropathologists according to ISUP protocols. A pathological lymph node assessment, either LNI- or LNI+, was given for each case as the gold standard.

Image segmentation and deep learning features extraction

Both the entire prostate gland and the primary tumor of PCa were delineated concurrently on PSMA PET/CT images as two separate labels of the volume of interest (VOI) by two nuclear medicine physicians (QK Ma and B Chen) using 3D Slicer (version 5.6.0, <https://www.slicer.org>). Following the segmentation of the VOIs, the PET and CT images underwent preprocessing procedures such as normalization and cropping to improve data quality and diversity. Then, the Med3D network model was utilized to extract deep learning features from PET and CT images. The Med3D network is pre-trained with heterogeneous datasets of large-scale three-dimensional medical imaging modalities (including CT and MRI) that contain distinctive anatomical regions (e.g. brain, prostate, liver, spleen, etc.), and it can be generalized and transferred as a feature extractor to other medical tasks with small datasets [34].

Development and validation of multimodal deep learning model based on PSMA PET/CT imaging

To establish a multimodal prediction model for LNI, we derived a multi-kernel support vector machine (SVM) classifier, which allowed for the integration of information from different modalities by kernel combination trick, thereby enhancing the accuracy and stability of classification. PET and CT features, clinical parameters and SUV_{max}

as independent predictors, were subsequently inputted into the classifier for the prediction and classification of LNI disease in PCa. The original problem was the following, when applying kernel fusion method in multi-kernel SVM:

$$\begin{aligned} \min_W \quad & \frac{1}{2} \sum_{m=1}^M \beta_m \|W^m\|^2 + C \sum_{i=1}^n \xi_i, \\ \text{s.t.} \quad & y_i \left(\sum_{m=1}^M \beta_m (W^m)^T \phi^m(x_i^m) + b \right) \geq 1 - \xi_i, \\ & \xi_i \geq 0, i = 1, 2, \dots, n. \end{aligned} \quad (1)$$

where n is the number of the participants, $y_i \in \{1, -1\}$ represents the label (either LNI- or LNI+) of the corresponding, i.e., i -th participant. The number of the adopted connection patterns is denoted as M . W^m denotes the hyper-plane in the reproducing kernel Hilbert space and β_m represents the combination weight of the m -th pattern. The entire pipeline of our model development procedure is depicted in Fig. 1. Furthermore, given the limited sample size, the leave-one-out cross-validation was utilized in the training cohort for model training and validation, in which the classifier was trained iteratively on all samples minus one, and the one held out for testing until all samples had served as a testing set. To confirm the stability of the deep learning model, a 5–5 leave out and 10–10 leave-out cross-validation were also performed.

In this study, predictors for model development included clinical parameters (initial PSA level, biopsy ISUP grade group and cT stage), SUV_{max} of the primary tumor and deep learning features extracted from the entire prostate on PSMA PET/CT images. Three models were developed: Model 1 (clinical parameters and SUV_{max} as input), Model 2 (only deep learning features as input), and Model 3 (clinical parameters, SUV_{max} and deep learning features together as

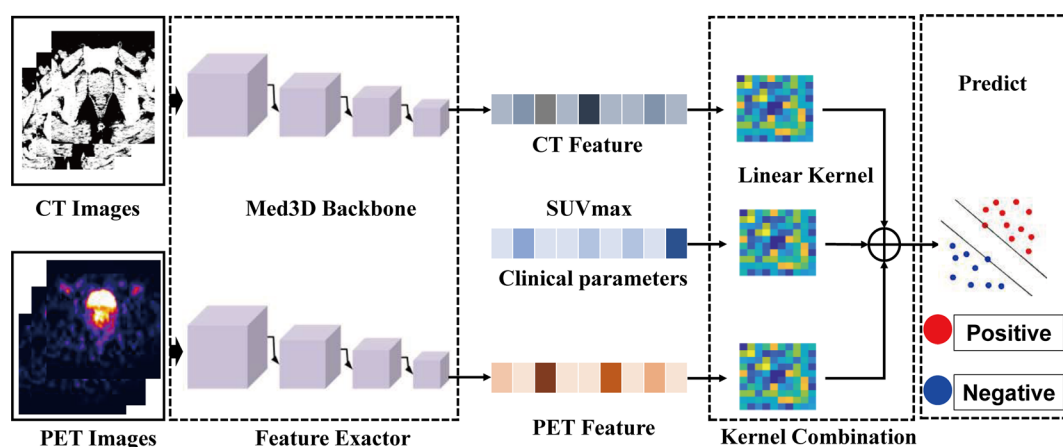


Fig. 1 The entire pipeline of the proposed multimodal deep learning model based on PSMA PET/CT imaging, SUV_{max} and clinical parameters for the prediction of lymph node invasion

input). The code has been uploaded to GitHub (<https://github.com/MoHuaZR/ProstatePrediction>).

Statistical analysis

In this study, categorical variables were analyzed using Fisher exact test and Mann Whitney *U* test, while continuous variables were compared using Student *t*-test and Mann Whitney *U* test. The discriminative ability of the models was evaluated by calculating the area under the receiver operating characteristic curve (AUC), and the AUC among models was compared using the Delong's test. The models included the PSMA PET/CT visual evaluation result, well-established nomograms, and multimodal deep learning models. Performance metrics such as accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were determined based on the optimal cutoff values identified by the maximum Youden index. Calibration curve and decision curve analysis (DCA) were employed to reflect the disagreement between predicted and observed probabilities, as well as clinical application value of different models. The calibration of our models was examined by the Hosmer-Lemeshow test. Statistical and graphical analyses were conducted using SPSS (version 26.0, IBM SPSS Statistics) and R program (version 4.3.1,

<https://www.r-project.org>). All tests were two-sided and *p* values < 0.05 were considered statistically significant.

Results

Baseline characteristics

Table 1 presents detailed information about the baseline characteristics of the study cohort. A total of 116 eligible patients were enrolled, of whom 82 were assigned to the training cohort and 34 were assigned to the test cohort. There were no significant differences in baseline characteristics between the training and test cohorts.

Final histopathological examination

In the postoperative pathological examination, 24 (29%) patients in the training cohort were identified as having LNI, while 11 (32%) patients in the test cohort were determined to have LNI. The LNI+ group had a higher percentage of ISUP grade 5 biopsies (17% vs. 46%, *p*=0.003; 13% vs. 64%, *p*=0.044), a higher prevalence of T3b or T4 disease (14% vs. 59%, *p*<0.001; 13% vs. 73%, *p*=0.007), and a higher occurrence of miN1 results (2% vs. 67%, *p*<0.001; 4% vs. 64%, *p*<0.001) in both cohorts. Besides, the LNI+ group

Table 1 Patient characteristics

Characteristics	Training cohort (<i>n</i> =82)			Test cohort (<i>n</i> =34)			<i>p</i>
	LNI-	LNI+	<i>p</i>	LNI-	LNI+	<i>p</i>	
No. (%)	58 (71)	24 (29)	NA	23 (68)	11 (32)	NA	NA
Age, mean (SD), years	66.6 (6.6)	66.1 (6.3)	0.745	68.0 (4.6)	65.7 (10.4)	0.494	0.315
Injected dose, median (IQR), MBq	158.7 (146.5–178.0)	168.2 (160.4–188.7)	0.023	148.0 (146.5–171.1)	148.0 (146.5–155.8)	0.532	0.058
Patients' height, mean (SD), cm	166.8 (5.6)	165.5 (5.1)	0.327	168.0 (4.1)	166.9 (4.3)	0.494	0.234
Patients' weight, mean (SD), kg	68.2 (9.0)	63.5 (7.6)	0.028	65.7 (8.4)	65.8 (8.7)	0.958	0.543
PSA, ng/ml			0.015			0.113	0.418
Median	17.4	31.3		19.8	38.4		
IQR	11.8–32.7	14.4–57.7		11.6–37.5	18.0–67.5		
ISUP grade group, <i>n</i> (%)			0.003			0.044	0.701
1	6 (10)	1 (4)		2 (9)	1 (9)		
2	9 (16)	0 (0)		4 (17)	1 (9)		
3	17 (29)	5 (21)		5 (22)	1 (9)		
4	16 (28)	7 (29)		9 (39)	1 (9)		
5	10 (17)	11 (46)		3 (13)	7 (64)		
Clinical T stage, <i>n</i> (%)			<0.001			0.007	0.230
T1c/T2	48 (83)	8 (33)		11 (48)	3 (27)		
T3a	2 (3)	2 (8)		9 (39)	0 (0)		
T3b/T4	8 (14)	14 (59)		3 (13)	2 (73)		
Tumor SUV _{max}			0.022			0.645	0.794
Median	13.7	16.7		13.8	12.7		
IQR	8.6–20.9	12.6–38.1		8.3–20.5	9.5–31.8		
PSMA PET/CT findings, <i>n</i> (%)			<0.001			<0.001	0.805
miN0	57 (98)	8 (33)		22 (96)	4 (36)		
miN1	1 (2)	16 (67)		1 (4)	7 (64)		

Table 2 Performance of the three models in our study for the prediction of lymph node metastasis

Model	AUC (95%CI)	Accuracy	Sensitivity	Specificity	PPV	NPV
Model 1 ^a	0.74 (0.62–0.86)	0.68	0.88	0.60	0.48	0.92
Model 2 ^b	0.86 (0.77–0.95)	0.87	0.71	0.93	0.81	0.89
Model 3 ^c	0.89 (0.81–0.97)	0.89	0.71	0.97	0.89	0.89

^aClinical parameters and SUV_{max} as input^bOnly deep learning features as input^cClinical parameters, SUV_{max} and deep learning features together as input**Table 3** Performance of the original preoperative nomograms, PET/CT visual evaluation result, and the proposed multimodal deep learning model for the prediction of lymph node invasion in the training cohort ($n=82$) and test cohort ($n=34$)

Data cohort	Method	AUC (95%CI)	Accuracy	Sensitivity	Specificity	PPV	NPV
Training	Multimodal model	0.89 (0.81–0.97)	0.89	0.71	0.97	0.89	0.89
	PSMA PET/CT visual evaluation	0.82 (0.73–0.92)	0.89	0.67	0.98	0.94	0.88
	MSKCC ^a	0.75 (0.61–0.89)	0.33	0.92	0.09	0.29	0.71
	Briganti 2017 ^b	0.73 (0.61–0.86)	0.45	0.92	0.26	0.34	0.88
Test	Multimodal model	0.85 (0.69–1.00)	0.88	0.73	0.96	0.89	0.88
	PSMA PET/CT visual evaluation	0.80 (0.64–0.95)	0.85	0.64	0.96	0.88	0.85
	MSKCC ^a	0.79 (0.61–0.97)	0.38	0.91	0.13	0.33	0.75
	Briganti 2017 ^b	0.69 (0.50–0.88)	0.47	0.91	0.26	0.37	0.86

^aAccuracy, sensitivity, specificity, PPV and NPV were calculated with a threshold probability of 5%^bAccuracy, sensitivity, specificity, PPV and NPV were calculated with a threshold probability of 7%

exhibited higher initial PSA levels (31.3 vs. 17.4 ng/ml, $p=0.015$) and SUV_{max} (16.7 vs. 13.7, $p=0.022$) compared to the LNI- group in the training cohort, while in the test cohort, there is no significant difference between these two metrics. In both cohorts, there was no significant difference in age between the LNI+ and LNI- groups (Table 1).

Model development and selection

In the internal leave-one-out cross-validation of our training cohort, Model 3 with multimodal predictors demonstrated the highest AUC of 0.89 (95% confidence interval [CI] 0.81–0.97) compared to the other two models (Model 1, 0.74, [95% CI 0.62–0.86]; Model 2, 0.86, [95% CI 0.77–0.95]) (Table 2). The results of the 5–5 leave-out and 10–10 leave-out cross-validation were similar to those of 1-split (Table S1–S2). Thus, the multimodal deep learning model based on preoperative PSMA PET/CT imaging showed potential for enhancing the prediction of pathological LNI and patient selection for ePLND over the other models. Subsequently, the multimodal model was applied to the test cohort for performance evaluation and further compared with commonly used nomograms and PSMA PET/CT visual evaluation result.

Performance, comparison, and clinical application of the multimodal deep learning model, PSMA PET/CT visual evaluation result and original nomograms in the training cohort ($n=82$)

As displayed in Table 3; Fig. 2A, the multimodal deep learning model demonstrated superior discriminative ability in predicting LNI compared to the visual evaluation result of PSMA PET/CT, MSKCC nomogram, and Briganti-2017 nomogram (AUC 0.89 [95% CI 0.81–0.97] vs. 0.82 [95% CI 0.73–0.92] vs. 0.75 [95% CI 0.61–0.89] and 0.73 [95% CI 0.61–0.86], respectively, DeLong's test: $p<0.05$). At the optimal threshold of 47% determined using the Youden index, the accuracy, sensitivity, specificity, PPV, and NPV of the proposed multimodal deep learning model were 0.89, 0.71, 0.97, 0.89 and 0.89, respectively. Comparable sensitivity showed that the multimodal deep learning model provided only modest improvement in LN staging over PSMA PET/CT visual evaluation result, which tended to underestimate the actual risk of LNI. Using a risk cutoff of 31% could maintain an enhanced diagnostic accuracy, with a sensitivity of 0.88 and a specificity of 0.76 to avoid unnecessary ePLND (Table S3).

In terms of calibration, the Hosmer-Lemeshow test yielded nonsignificant statistics ($p>0.05$) for our multimodal model and clinical nomograms, indicating a high calibration accuracy. The p values of our multimodal model, the MSKCC and Briganti-2017 nomograms were 0.865, 0.896 and 0.927, respectively. Although there is an underestimation of LNI risk when the predicted risk is between 10%

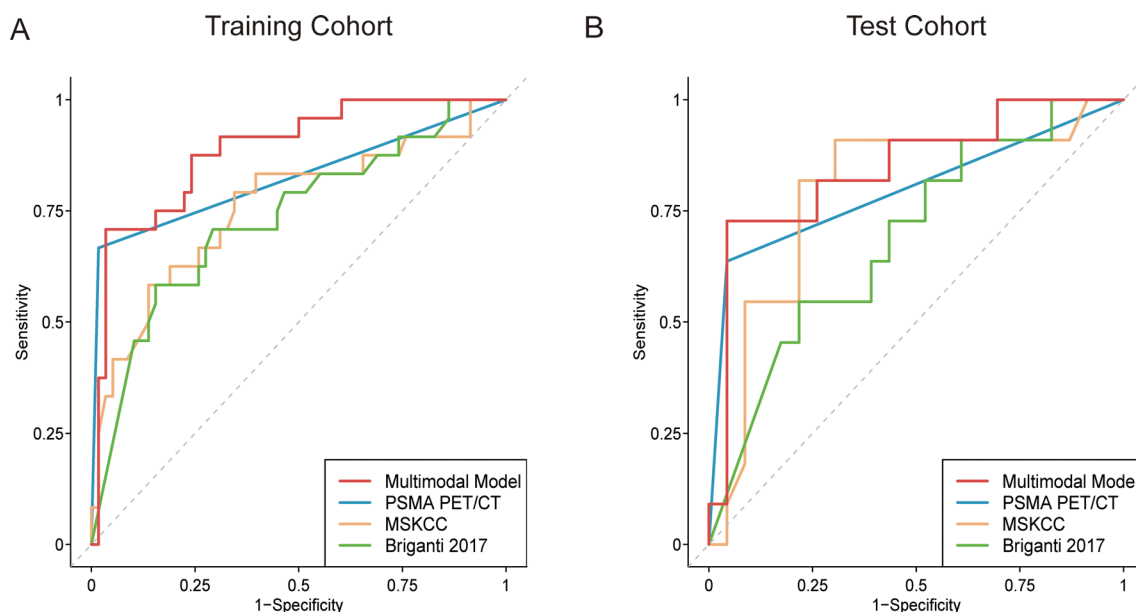


Fig. 2 ROC curves for commonly used nomograms, PSMA PET visual evaluation result, and the proposed multimodal deep learning model based on PSMA PET/CT features, SUV_{max} and clinical parameters. The multimodal model yielded the highest AUC, followed by PSMA

PET visual evaluation result, the MSKCC nomogram and the Briganti-2017 nomogram in the training cohort (AUC 0.89 vs. 0.82 vs. 0.75 vs. 0.73) (A) and the test cohort (AUC 0.85 vs. 0.80 vs. 0.79 vs. 0.69) (B)

and 25%, and an overestimation of LNI probability when the predicted risk is between 25% and 45%, as indicated in the calibration curve (Fig. 3C), our proposed model demonstrated good calibration, particularly for predicted risks over 45%, and was found to be comparable to the MSKCC (Fig. 3A) and Briganti-2017 nomograms (Fig. 3B). On DCA (Fig. 4A), the multimodal model reached an obvious higher net benefit than frequently used nomograms, treat-all and treat-none schemes at the risk threshold of $\geq 4\%$.

Regarding the decision support for ePLND using the common clinical nomograms, the MSKCC and Briganti-2017 nomograms would result in 53/82 (65%), and 43/82 (52%) unnecessary ePLND at a risk threshold of 5% and 7%, respectively, with a risk of missing only 2/82 (2%) LNI+patients (Table 4). Conversely, when the proposed multimodal deep learning model (with a cut-off value of 31%) and PSMA PET/CT visual evaluation result were used for candidate selection, 43/82 (52%) and 57/82 (70%) of ePLND could be avoided in our cohort, respectively, with only 3/82 (4%) and 8/82 (10%) LNI disease missed, respectively (Table 4). Of note, among the 24 LNI+patients, 8 of them had a negative PSMA PET/CT visual evaluation result, and our multimodal model accurately staged 5 out of the 8 misdiagnosed patients.

Performance, comparison, and clinical application of the multimodal deep learning model, PSMA PET/CT visual evaluation result and original nomograms in the test cohort ($n = 34$)

The proposed model showed significantly higher discrimination ability (AUC 0.85 [95% CI 0.69–1.00]) compared to the visual evaluation result of PSMA PET/CT (AUC 0.80 [95% CI 0.64–0.95], MSKCC nomogram (AUC 0.79 [95% CI 0.61–0.97]), and Briganti-2017 nomogram (AUC 0.69 [95% CI 0.50–0.88]) (Table 3; Fig. 2B). The accuracy, sensitivity, specificity, PPV and NPV of the proposed multimodal deep learning model were 0.88, 0.73, 0.96, 0.89 and 0.88, respectively, which was consistent with the performance in the training cohort.

Regarding the calibration, the p values of our multimodal model, the MSKCC and Briganti-2017 nomograms were 0.943, 0.994 and 0.893, respectively. Overall, our proposed model (Fig. 3F) demonstrated good calibration comparable to the MSKCC (Fig. 3D) and Briganti-2017 nomograms (Fig. 3E). On DCA (Fig. 4B), the multimodal model also demonstrated higher net benefit within a wide range of risk threshold.

As for the clinical implication value, MSKCC and Briganti-2017 nomograms would result in 20/34 (59%), and 17/34 (50%) unnecessary ePLND, respectively, with only a 1/34 (3%) risk of LNI+patients missed (Table 4). Conversely, the proposed multimodal deep learning model (with a cut off value of 31%) and PSMA PET/CT visual

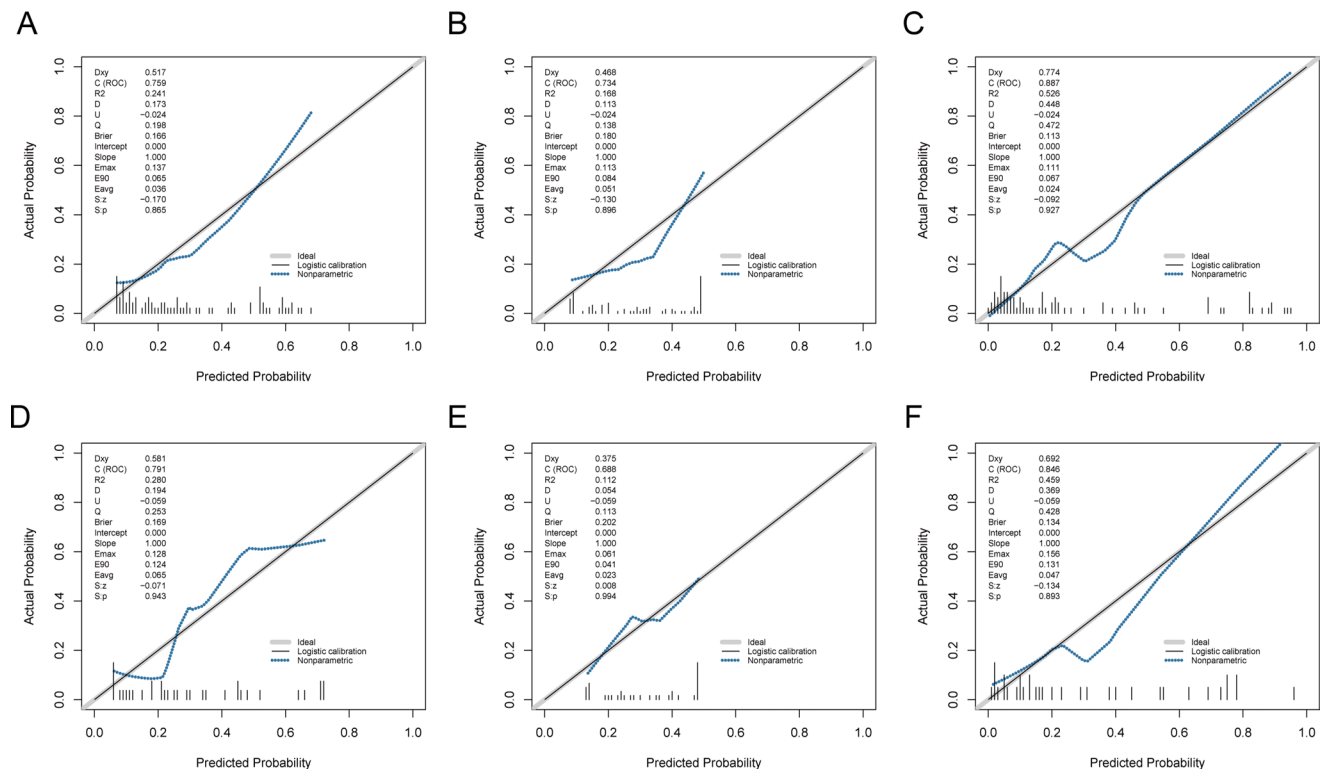


Fig. 3 Calibration plot of predicted probability versus observed probability of lymph node involvement for the MSKCC nomogram (A, D), the Briganti-2017 nomogram (B, E) and the proposed multimodal model (C, F) in the training cohort ($n=82$) and test cohort ($n=34$)

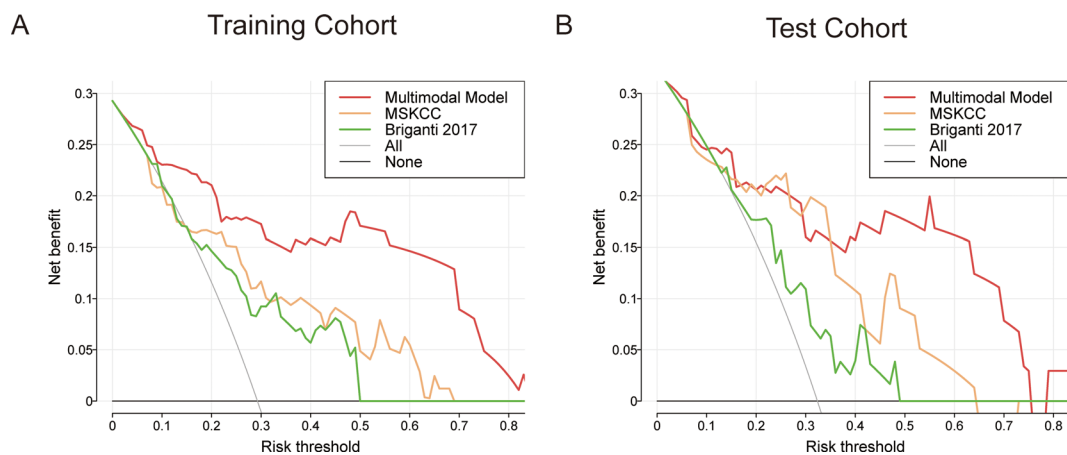


Fig. 4 Decision curve analysis (DCA) of the proposed multimodal deep learning model compared to conventional nomograms, treat-all and treat-none schemes in the training cohort ($n=82$) (A) and test

cohort ($n=34$) (B). The DCA depicts the net benefit (NB, y-axis) of a model or a strategy according to a risk threshold (x-axis)

evaluation could result in 17/34 (50%) and 22/34 (65%) of omitted ePLND, with only 3/34 (9%) and 4/34 (12%) LNI disease missed, respectively (Table 4). Notably, among 4 LNI+ patients with a negative PSMA PET/CT visual evaluation, our multimodal model accurately staged 1 patient, and 3/4 of the patients were misdiagnosed.

Discussion

Final histopathological examination following ePLND is considered the gold standard for N-staging of clinically significant localized PCa. As the overall incidence of lymph node metastasis is relatively low and this surgical intervention is not free of complications, it is intuitively mandatory to select patients who are best suited to undergo this diagnostic surgical intervention and in whom the benefit is

Table 4 Clinical implications of the nomograms available, PSMA PET/CT visual evaluation results and the proposed multimodal model in the training cohort ($n=82$) and test cohort ($n=34$)

Data cohort	Method	Below the cutoff (PLND not recommended), n (%)			Equal to or above the cutoff (PLND recommended), n (%)		
		Total	Without LNI	With LNI	Total	Without LNI	With LNI
Training	Multimodal model ^a	46 (56)	43 (93)	3 (7)	36 (44)	15 (34)	21 (48)
	PSMA PET/CT visual evaluation ^b	65 (79)	57 (88)	8 (12)	17 (21)	1 (6)	16 (94)
	MSKCC ^c	7 (9)	5 (71)	2 (29)	75 (91)	53 (71)	22 (29)
	Briganti 2017 ^d	17 (21)	15 (88)	2 (12)	65 (79)	43 (66)	22 (34)
Test	Multimodal model ^a	20 (59)	17 (85)	3 (15)	14 (41)	6 (43)	8 (57)
	PSMA PET/CT visual evaluation ^b	26 (76)	22 (85)	4 (15)	8 (24)	1 (13)	7 (87)
	MSKCC ^c	4 (12)	3 (75)	1 (25)	30 (88)	20 (67)	10 (33)
	Briganti 2017 ^d	7 (21)	6 (86)	1 (14)	27 (79)	17 (63)	10 (37)

^a31% cutoff^bNo cutoff^c5% cutoff^d7% cutoff

highest. In this study, we developed and tested a multimodal deep learning prediction model that incorporates clinical parameters, quantitative information from PSMA PET, and features extracted from entire prostate imaging on PSMA PET/CT to predict LNI at RP with ePLND as standard reference. We have also assessed the performance of our model against conventional prediction models. Our results suggest that a model including deep learning features or/and parameters from PSMA PET scans could enhance the accuracy of LNI prediction and might improve patient selection for ePLND beyond what can be achieved with clinical parameters alone. In our cohort, the proposed multimodal prediction model outperformed both PSMA PET/CT visual evaluation results and established preoperative models such as the Briganti-2017 and MSKCC nomograms.

Our findings suggest that the current nomograms for predicting LNI may be suboptimal for preoperative staging of PCa patients without PSMA PET/CT, with a PPV of approximately 37%. Our results align with recent research indicating that the MSKCC and Briganti-2017 nomograms have demonstrated relatively low accuracy in predicting LNI, with an AUC around 0.7 [35], as well as high sensitivity and low specificity, resulting in unnecessary ePLND for a significant number of individuals while missing only a few LNI cases [29]. The decision support provided by common clinical nomograms like the MSKCC and Briganti-2017 nomograms would lead to more than half of unnecessary ePLND with a risk threshold of 5% and 7%, respectively. Patients who underwent unnecessary ePLND may have more complications, longer operative room time and higher costs. On average, performing ePLND can increase the operative time of a standard RP by 45 to 90 min compared with RP without pelvic lymph node dissection. Omitting futile ePLND can be expected to bring further benefits by avoiding additional costs associated with histopathological analysis and side

effects such as lymphoceles, lymphedema, and thromboembolic events. The avoidance of overtreatment and reduction in the number of unnecessary ePLND procedures, while ensuring that no additional LNI cases are missed, is currently a critical unmet need in the management of PCa.

PSMA PET/CT is becoming more commonly used for the initial staging of patients with PCa due to its improved diagnostic precision in detecting metastases compared to conventional imaging techniques, particularly for individuals with intermediate- or high-risk PCa. However, the limited spatial resolution of PET may lead to challenges in detecting micro-metastasis, potentially contributing to the existing suboptimal sensitivity levels [10]. Our results showed that the sensitivity of PSMA PET for evaluating lymph node metastasis was less than 70%, which might be related to the fact that all patients included in our study were candidates for ePLND. Recent studies examining the effectiveness of PSMA PET/CT in detecting LNI indicated its high specificity and a sensitivity comparable to nomograms [36, 37]. Unfortunately, as of now, PSMA PET/CT has not been incorporated into the clinical nomograms. It is believed that the combination of predictors from PSMA PET/CT and clinical metrics may be valuable in predicting LNI, particularly due to its high negative predictive value [29, 38] (89% in our training cohort and 88% in our test cohort) in assessing nodal metastases. This, in one aspect, may help to reduce overtreatment to some extent. Furthermore, integrating deep learning and molecular imaging to assess PCa with multi-focal features may be a potential tool for predicting lymph node metastasis.

Our multimodal approach may provide a new alternative for balancing sensitivity and specificity, addressing the clinical dilemma by sparing over a half of unnecessary PLND, while missing only a few positive lymph nodes. Our data suggests that adoption of multimodal predictors, including

deep learning features from whole prostate or/and primary tumor images, might improve the sensitivity in detecting LNI and reduce the number of false-negative results in PSMA PET scans. Recently, there have been frequent reports on the added value of primary tumor characteristics from PSMA PET and mpMRI, such as PSMA-positive volume, total PSMA accumulation, maximum diameter of the index lesions, and radiological T stage on mpMRI, in predicting LNI in PCa patients [18, 29, 39, 40]. Moreover, Cysouw et al. found radiomics features of the primary tumor on PSMA PET scan could effectively reflect the expression level and spatial distribution of PSMA. These features demonstrated a discrimination ability that was equal to or greater than that of standard PET metrics [41]. In contrast to the traditional radiomics method, deep learning algorithms extract abstract and informative features rather than relying on explicit pre-defined features, making them more adept at addressing clinical challenges [22]. Therefore, we employed the Med3D network to extract features from PSMA PET/CT images, which turned out to significantly improve the performance of the model only based on clinical parameters in our study. On the other hand, from a clinical standpoint, our findings validate the use of a PSMA PET/CT-based multimodal deep learning model in some conjunction with MSKCC and/or Briganti nomograms, which can accurately predict LNI and facilitate the selection of patient for ePLND, as it significantly improved the discrimination in the test cohort.

Despite several strengths, our study has multiple limitations. Firstly, the small sample size and retrospective design may introduce selection bias. Secondly, the training cohort and test cohort were from the same medical center, and future external test in a larger and more diverse patient population across multiple medical centers is warranted. However, the multimodal model, which incorporates deep learning features, outperformed the clinical models based solely on data from our cohort. This indicates that the inclusion of PSMA PET/CT deep learning features enhances the accuracy of predicting LNI compared to relying solely on clinical information. Thirdly, since many patients did not undergo MRI-targeted biopsy and MRI data was missing in some cases, we did not include MRI parameters in our analysis or compare our model with the Briganti-2019 nomogram. We acknowledge this limitation and plan to address it in future studies by including more patients. Additionally, the study was conducted using only one available PSMA tracer. The performance of the deep learning tools should be assessed in PET/CT scans utilizing other PSMA tracers and across studies conducted at various hospitals.

Conclusion

Our multimodal deep learning model, integrating clinical information, SUV_{max} and PSMA PET/CT deep learning features of the prostate, outperforms visual lymph node staging on PSMA PET/CT, as well as the MSKCC and Briganti-2017 nomograms in accurately predicting LNI in patients with clinically localized PCa. While further external validation is necessary, it has the potential to minimize the risk of unnecessary ePLND procedures without missing cases of LNI.

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Author contributions All authors contributed to the study conception and design. Material preparation, data collection, and analysis were performed by Qiaoke Ma, Bei Chen, Robert Seifert, Ling Xiao, Jinhui Yang, Wei kai Li, Yongxiang Tang, and Shuo Hu. The deep learning network and multimodal models were built by Rui Zhou and Weikai Li. The first draft of the manuscript was written by Qiaoke Ma, and Bei Chen. Robert Seifert, Axel Rominger, Kuangyu Shi, Yongxiang Tang and Shuo Hu revised the work critically for important intellectual content. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate This retrospective study was approved by the Institutional Review Board of Xiangya Hospital, Central South University and the requirement for written informed consent was waived.

Consent for publication Not applicable.

Competing interests Robert Seifert has received research/travel support from the Boehringer Ingelheim Fund and the Else Kröner-Fresenius-Stiftung, as well as travel support and lecture fees from Novartis

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Authors and Affiliations

Qiaoke Ma¹ · Bei Chen¹ · Robert Seifert² · Rui Zhou³ · Ling Xiao¹ · Jinhui Yang¹ · Axel Rominger² · Kuangyu Shi^{2,4} · Weikai Li⁵ · Yongxiang Tang^{1,2,6} · Shuo Hu^{1,6,7} 

✉ Weikai Li
leeweikai@cqjtu.edu.cn

✉ Yongxiang Tang
xyyf0401@qq.com

✉ Shuo Hu
hushuo2018@163.com

¹ Department of Nuclear Medicine, Xiangya Hospital, Central South University, No. 87 Xiangya Road, Changsha, Hunan 410008, P.R. China

² Department of Nuclear Medicine, Bern University Hospital, Inselspital, University of Bern, Bern, Switzerland

³ School of Zhang Jian, Nantong University, Nantong, Jiangsu, P.R. China

⁴ Department of Informatics, Technische Universität München, Munich, Germany

⁵ College of Mathematics and Statistics, Chongqing Jiaotong University, Chongqing, P.R. China

⁶ National Clinical Research Center for Geriatric Disorders (Xiangya), Changsha, Hunan, P.R. China

⁷ Key Laboratory of Biological Nanotechnology of National Health Commission, Changsha, Hunan, P.R. China